Qian Ouyang

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Education

Xi'an Jiaotong University 2023 - 2027 (expected)
Undergraduate studies in Artificial Intelligence

Artificial Intelligence Experimental Class

University of Notre Dame August - December 2025
Undergraduate Exchange Student (non-degree)

Semester GPA: 3.917/4.0 | 12 credits

Selected coursework: Computer Vision - graduate level (A); Natural Language Processing (A); Computer Graphics (A).

Research Experience

University of Notre Dame

Research Assistant October 2025 - Present
Prof. Tengfei Luo's group | Department of Aerospace and Mechanical Engineering

Machine learning for polymer simulation

Investigated machine-learning interatomic potentials for polymer-property calculations. Initial work with UMA / FAIR-Chem explored finite-strain calculations of polymer elastic response, with GPU-memory constraints limiting the simulations.

With MACE, evaluated preliminary polymer-density estimates from molecular dynamics at 300 K following classical structure preparation. Compared the estimates with reference data and examined sampling and differences in physical conditions when interpreting the results. Also compared GPU memory use, runtime, and GPU-hours across selected MACE runs to assess the computational cost of these calculations.

Representation learning for molecular dynamics

Investigated how molecular representations affect physical prediction in coarse-grained polymers. Compared inputs describing local packing and chain geometry, and evaluated temporal-learning models against persistence and linear autoregressive baselines.

Moved to atomistic water to separate the effects of input information from the training objective. Used pretrained MACE features and atomic velocities for short-time displacement prediction, comparing temporal JEPA training with direct physical supervision. Evaluated the learned representations through the same downstream prediction task, examining whether improvements in the training objective translated into lower displacement errors.

Summer Research Participant (iSURE) July - September 2026

Course Project

Driving Coach: Lane and Traffic-Object Detection Fall 2025
Computer Vision, University of Notre Dame | Instructor: Prof. Adam Czajka

Adapted an existing YOLOv1-style detector with a pretrained ResNet-18 backbone for lane and traffic-object detection in BDD100K driving scenes. Used an unseen Shenzhen highway video for qualitative evaluation and analyzed errors involving reflections, lane geometry, and traffic signs.

Skills

Programming: Python, C++